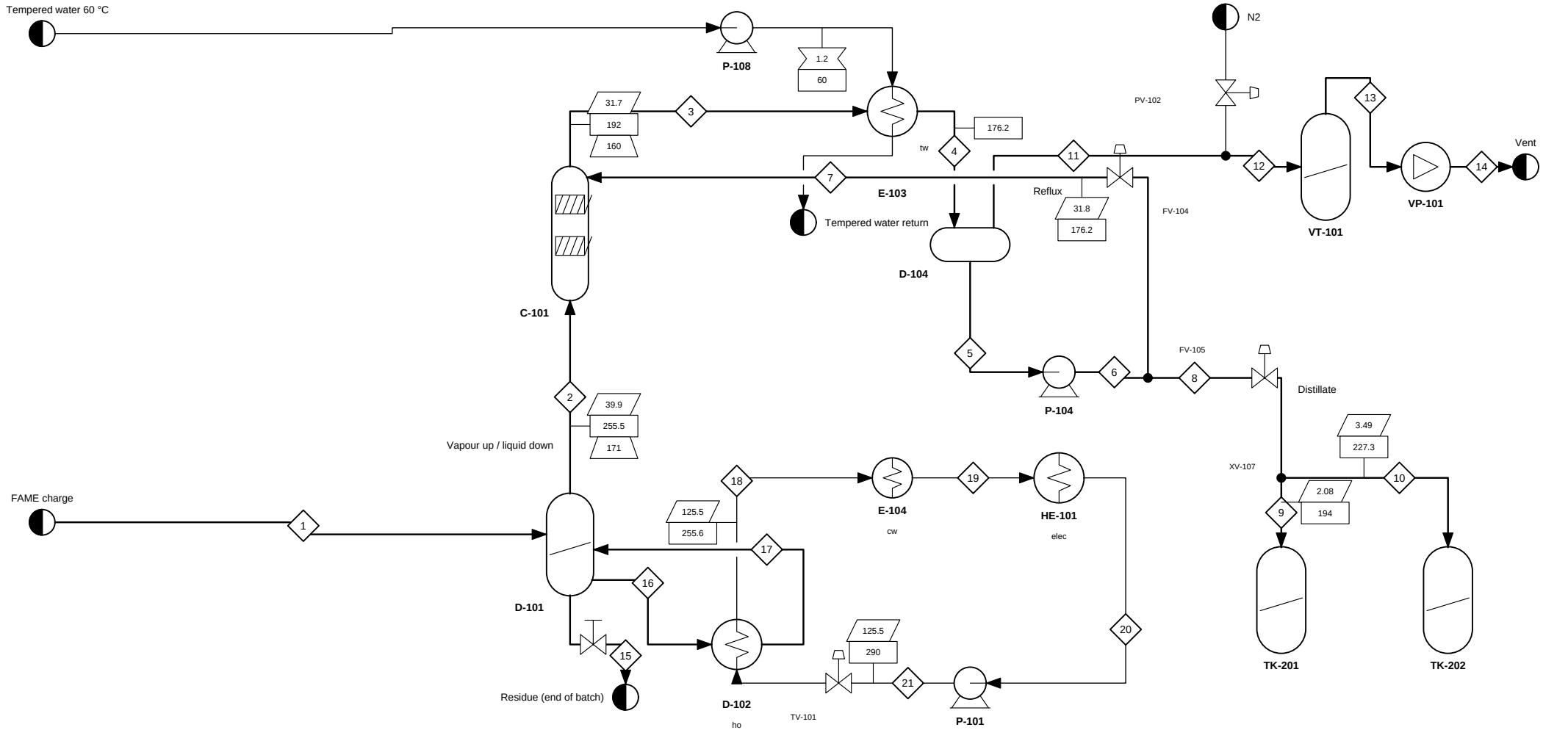


HE-101 Hot-Oil Electric Heater 12 kW	P-101 Hot-Oil Pump	E-104 Hot-Oil Cooler (cool-down)	D-102 Reboiler (hot oil)	D-101 Batch Still (Pot) 150 L	C-101 Packed Column, 20 Stages	P-108 Tempered-Water Pump	E-103 Overhead Condenser	D-104 Reflux Drum 4 L	P-104 Reflux / Distillate Pump	VT-101 Vacuum Trap	VP-101 Vacuum Pump	TK-201 Off-Cut Receiver	TK-202 Product Receiver
--	-----------------------	--	-----------------------------	-------------------------------------	--------------------------------------	------------------------------	-----------------------------	--------------------------	--------------------------------------	-----------------------	-----------------------	----------------------------	----------------------------



NOTES

1. Flags: simulator model, noise off. Start of campaign at total reflux, except 9 (off-cut, 120 min) and 10 (product cut, 547 min).

2. Flows kg/h (water m3/h); temperatures °C; pressures mbar abs. Sizes indicative.

3. Batch charge 110 L (95.7 kg) FAME; residue stays in D-101 until discharged.

Mass Flow (kg/h)	Volume Flow (m3/h)	Temperature (°C)	Pressure (mbar abs)	Process Line	Utility Line
Block Valve	Control Valve	Process Input / Output	Stream Number	Line Crossing	Heat Exchanger
Centrifugal Pump	Vessel / Drum	Packed Column	Separator	Blower / Fan	

VACUUM BATCH DISTILLATION VER 1 - OPERATOR TRAINING SIMULATOR			
PROCESS FLOW DIAGRAM			
VACUUM BATCH DISTILLATION C-101 - METHYL PALMITATE CAMPAIGN			
DRAWING No. VBD-PFD-100	REV. 0	SIZE A4	SHEET 1 OF 1
DRAWN Apinan Soottitantawat	CHECKED	SCALE NTS	DATE 2026-10-09